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Describe the locus of z , and give details of the loci features.

$$\left| \frac{z+1}{z-1} \right| = 2$$

Find all exact solutions (in rectangular form) in terms of a , of the equation

$$(z-a)^3 = 8$$

If $|z| = 1$, where $z = x + yi$, show that the following expression is purely imaginary

$$z^n - z^{-n}$$

Justify your answer.

Given that

$$\arg\left(z - \frac{uw}{2}\right) = \frac{\pi}{4}$$

describe and draw the locus of points that represent the complex number z .

The roots of the cubic equation, $2x^3 + px - (3 + 5i)x + q = 0$, are $x \in \{i, \frac{1}{k}, -1 - k\}$.

Use this information to find p, q, k where $k \in \mathbb{R}$.

Prove

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

is true for all complex numbers z_1 and z_2 .

Given

$$w^2 = 4\text{cis}\left(\frac{\pi}{4}\right)$$

find the value(s) of w^5 in polar form.

Prove

$$\overline{z^n} = (\overline{z})^n$$

Find the value of b if

$$\left| \frac{1+bi}{3-4i} \right| = 1$$

Let $z = a + bi$.

If it is known that $a^2 = 3b^2$, show that the following equation is purely imaginary

$$\frac{z^2}{\overline{z}}$$

Given

$$2|z - 2 - i| - |z + i| = 0$$

FULLY describe the properties of the complex number z .

If

$$z^6 = 1 + i$$

find ALL solutions for z , s.t.

$$-\frac{\pi}{6} \leq \arg(z) \leq \pi$$

Show that

$$z^p = 2 + 3i$$

and

$$z^q = 1 - i$$

have no common solutions given $p, q \in \mathbb{N}$

Hint: You do not have to compare arguments.

Describe fully and give the equation of the points z which satisfy the equation

$$|z - 2 - i| = |z + 3|$$

If $a, b > 0$, $a, b \in \mathbb{R}$, s.t.

$$\sqrt{\frac{28 + 6\sqrt{3}}{9}} = a + b\sqrt{3}$$

find the values of a and b .

Given that $z = x + yi$ where $x, y \in \mathbb{R}$, is a point on the curve $|z - a| = k$, where $a, k > 0$, $a, k \in \mathbb{R}$, then prove $\bar{z}$ the complex conjugate of z also lies on the curve.

Solve the equation

$$\sqrt{x + k} + \sqrt{x - k} = k$$

where $k > 0$, $k \in \mathbb{R}$.

Give your answer in terms of k then find the value of k if the solution is k .

Prove that if

$$z^2 + \frac{1}{z^2} = 2i$$

then

$$z + \frac{1}{z} = \pm \sqrt{2\sqrt{2}} \operatorname{cis} \frac{\pi}{4}$$

Given that $z = u + vi$, where $u, v \neq 0$, $u, v \in \mathbb{R}$, is a point on the line

$$|z - a| = |z - ai|$$

where $a > 0$, $a \in \mathbb{R}$, prove $\bar{z}$ the complex conjugate of z does not lie on this line.

Find ALL solutions to

$$(w^2 + 2)^2 = -3$$

using properties of complex numbers and algebraic techniques.

If $z = a + bi$, where $a, b \in \mathbb{R}$ and

$$\frac{1}{z} = \frac{\frac{a+b}{2} - 3i}{k^2}$$

after rationalizing, then find z and k .

If $v = 3 + ki$ and $w = 2 + ki$, find k if

$$\arg(u \cdot v) = -\frac{\pi}{4}$$

Describe fully and give the equation of the points z which satisfy the equation

$$|z + 7 - 4i| = 3|z - 1 + 4i|$$

Write your answer in the form

$$(x + p)^2 + (y + q)^2 = k$$

Solve the equation

$$\frac{\sqrt{x - k} + \sqrt{x + k}}{\sqrt{x - k} - \sqrt{x + k}} = \frac{1}{4}$$

for x in terms of k .

If $p = 4 + 2ai$ and $q = 3 - ai$, find the value of a if

$$\arg\left(\frac{p}{q}\right) = \frac{3\pi}{4}$$

Given

$$\left|\frac{z + 1}{z - 1}\right| = 2$$

where $z \in \mathbb{C}$, find the Cartesian equation of the locus and fully describe the features.

If $u = 4 + ki$ and $v = 6 - ki$, find k if

$$\arg\left(\frac{u}{v}\right) = -\frac{3\pi}{4}$$

Find and fully describe the locus where

$$|z - 2i| = 3|z + 1|$$

Given that

$$\sqrt{\frac{x}{1 - x}} = \sqrt{\frac{1 - x}{x}} = \frac{k}{4}$$

show that

$$x = \frac{1}{2} \pm \frac{\sqrt{k^2 - 64}}{2k}$$
